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By e-mail only:

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Dear Mr Eggleton,

S/COND/21/1998, Discharge of Condition 6 of S/OUT/13/1555: Phasing
S/COND/22/0031, Discharge of Condition 8 of S/OUT/13/1555: Strategic Design Code
Rowborough and South Marston Villages, Bus Operators' Consultation Response

1. Background

I am writing on behalf of both Stagecoach West and Go South Coast (trading as Swindon's Bus Company), as the two commercial bus operators in the Borough. Both companies operate commercial bus services through or in close proximity to the New Eastern Villages (NEV) allocation. Neither directly serves South Marston, or the Rowborough site. Both lie is off the current public transport network, and largely severed from the wider built form of Swindon by the A419, and the rest of NEV by the Great Western Main Line. For this and a number of other reasons, it is unclear which company will ultimately offer services to South Marston and Rowborough. However, this advice is applicable to any bus operator, and in so doing we hope to ensure that the Council's aspirations for public transport at Rowborough, and within the wider NEV, are achieved as far as they can be.

These aspirations are set out in diverse places in adopted planning policy for NEV. This is unusually comprehensive, and includes not only the Local Plan Core Strategy 2026, wherein NEV is the largest single allocation, but also several Supplementary Planning Documents (SPD) that also form part of the statutory development plan for the area. In particular, the Local Plan needs to achieve a substantially higher bus mode share from all its strategic allocations, including NEV, than for existing neighbourhoods in Swindon, mainly to satisfactorily mitigate transport impacts within foreseeably deliverable highways capacity. However, since the adoption of the Local Plan and SPDs, a legally binding national strategy to achieve net zero carbon by 2050 demands a substantial reduction in personal car use to be met.

It has to be said that, thus far, the experience with strategic allocations elsewhere in Swindon, including Wichelstowe and Tadpole Garden Village, suggests that approaches taken in the recent past are entirely failing to meet these objectives. It is essential that the implementation of development at NEV achieves a transformational approach to sustainable modes, and bus services in particular.

Therefore, we appreciate being consulted on this important application to discharge matters specified in Condition 8 of the outline permission for South Marston Village Extension and Rowborough. For conciseness, we will continue to refer to these as "Rowborough", within this response. These relate to Phasing, including that of key infrastructure and bus service delivery; and to key design parameters set in a submitted design code for the initial Neighbourhood Area, to be known as Wanborough Green. The successful delivery of high-quality public transport to this important part of New Eastern Villages depends very much on these matters being properly considered at this stage.

We recognise that this is not a planning application and is a technical consultation. Stagecoach and Go South Coast (trading as Swindon's Bus Company) made representations to both outline planning applications for this development. Significant input was offered at that stage. The principle of development and many key matters are now established. This is nevertheless a vital and welcome opportunity to work through some of the salient details pursuant to the consent. It also allows us in some cases, to address questions and concerns that we raised prior to consent.

The Council is very dependent on the propitious initiation of delivery of homes on this site to meet its identified development needs and support the rate of development required in the Local Plan Core Strategy development trajectory. The development also need to deliver at scale and pace. A suitably fast, and rational development programme is highly supportive of the early delivery of bus services to residents, as soon as possible after first occupations take place.

The site is situated in a way that it is physically set apart from the existing town and equally, from the existing extent of the bus network run by either operator. The severance created by the A419 trunk road reinforces this, also making it much more likely that motorised modes will be used. Being unable to take advantage of existing facilities and services, the level of self-containment of the site will be low until key points in time at which on-site provision is made, most importantly, primary schools and convenience retail. This creates a substantially elevated risk that exceptionally high levels of car dependence will be established on early phases. The impact of these across the local network on traffic conditions and the efficient reliable operation of the wider bus network in the town is and remains a key concern of both operators.

Policies DE1 and NC3 of the adopted Local Plan 2026 expect the Strategic Design Code to cover, among a number of other things, "Any alterations to public transport routes to accommodate the defined phases of development within the site".

Condition 6 requires that the **phasing of delivery of bus service extensions and alterations to serve the development should be submitted** for determination at this stage.

However, we will set out the measures that would be necessary to rationally extend a bus service into the development, that lie within the control of the applicant to deliver.

Condition 8 requires that a Strategic Design Code is prepared and agreed to set key design paramters and inform the preparation of more detailed design guidance and reserved matters across the development.

2. Condition 6: The development phasing parameters

Condition 6 covers the phasing across the site including, specifically:

b) Major distributor roads/routes within the site, including a defined hierarchy of the road network;

c) Phased access strategy delivery and associated phased housing delivery;

d) Pedestrian / cycle connectivity to committed and emerging parcels of development within the New Eastern Villages;

From a bus operator perspective these are all critical to manner and timing of delivery of the public transport offer throughout the build-out as well as post-completion. The sustainability of the bus service in the lon term is highly dependent on phasing, since failure to provide a relevant bus offer to residents at the time of occupation will divert the potential bus mode share into car, in particular. The key elements from the perspective of bus provision are therefore the points at which the primary streets are provided.

The development phasing, including the Infrastructure Phasing Plan, is tied to occupation and phase triggers that are specified in the outline planning permission, and are thus fixed.

We recognise the need, expressed in section 1.5, for the agreed strategy to be effective on an on-going basis, *"in co-ordinating development across a number of phases to avoid piecemeal provision of disparate*

and underutilised infrastructure". In so doing it is nevertheless important that appropriate sequencing of key links intended to afford buses access to the site is achieved in a rational and timely manner.

It must be stressed that the site is not served by any public transport in the near vicinity. The built form takes the shape of two discrete vilages, one an extension of South Marston. Both stand off from the rest of NEV, being further severed from the Great Stall Neighbourhood by both the mainline railway and the A420.

In particular, we note that critical connectivity across the Great Western Main Line to effect offsite connectivity, demands two substantial new rail crossings.

- The "Green Bridge" to the west, will provide links for active travel modes only to the only pair of stops on the A420 east of Whalebridge and the employment area beyond.
- A new underpass is intended to accommodate a second all-vehicular and all-modes access point that will be geometrically unconstrained, and links to the A420 just west of Acorn Bridge, at a proposed 4-way signalised junction that combines the Park and Ride access and the easternmost access into Great Stall East. This site is also intended to be the terminal point of the main east-west public transport spine through NEV.

This severance from the reminder of NEV to the south is exacerbated by the near total lack of permeability across strips of third party land on the southern side of the A420.

While a frequent bus service is provided on the A420 by the Stagecoach S6 service there is only a single pair of bus stops available recently provided at the Panattoni Park junction (former Symmetry Park).

To conclude, it is not credible to assume that most residents could take advantage of any existing bus services, especially in the Rowborough Village, before the point that buses can penetrate the development, and appropriately pass through or terminate.

In the short-medium term, then, the development is very highly dependent on the South Marston Primary Street, and Old Vicarage Lane. The SDC recognises that *"6.2.6 Old Vicarage Lane provides one of the key accesses into the site from the A420, and aligns with one of the key desire lines connecting South Marston and the new development to the other North Eastern Villages, and in particular to the Secondary School that will be provided as part of the Great Stall East development south of the A420."* The existing accommodation structure beneath the railway dates back to Victorian times. As such, it is incapable of readily accommodating high level of vehicular traffic, and pedestrians and cycles.

For the avoidance of doubt we are of the view that the Vicarage Lane under-bridge does not provide a safe and suitable access point to public passenger-carrying vehicles due to its width and height.

In addition:

- we had always thus far anticipated that the loop defined as Secondary Street Phase B would, in fact, be a primary road and accommodate a bus route.
- We have never really understood the reason for two parallel primary streets between Rowborough and Old Vicarage Lane, one of which seems to us to be redundant. To run a bus terminal loop that largely passes through undeveloped land, using both these links, is entirely irrational and will fail to provide a reasonably accessible bus route to the majority of residents.

We see that 4 phases are specified.

Phase 1	South Marston Extension East 260 dwellings
Phase 2	Rowborough Central dwellings 1470 dwellings
Phase 3	Rowborough East 410 dwellings
Phase 4	South Marston Extension East 240 dwellings

We note that *"The phases are likely to overlap once development becomes established and could run in parallel to ensure delivery of required infrastructure in line with the respective delivery triggers…"* (3.5). We also see that comments are made suggesting strategic and primary infrastructure will be delivered on a consortium basis (3.9-3.10), which if our reading reflects an accurate understanding, would be welcome.

While we entirely understand that the annualised delivery of homes is not something that can be precisely predicted, we see that a steady 200 dwellings per annum is assumed in the phasing plan. It is very likely that the Hallam Land control will pass into the hands of either one larger developer, which would be likely to simultaneously operate from at least two outlets for much of the duration of the build, probably under two distinct differentiated brand propositions; or alternatively to a consortium of two or three builders each trading from its own branded outlet. The Taylor Wimpey control again would be likely to build out with at least two outlets in line with their typical practice, readily observable in recent years in Swindon and beyond. Hannicks will dispose of their interest, which is smaller and focused on South Marston, and it is likely that a further set of "developer flags" will be associated with this. Therefore, at least 4 outlets are likely to be trading at any one time, once the development has fully mobilised. We also note and recognise that the submission implies that multiple development "fronts" are likely to arise, especially later in the development trajectory, when the Eastern Access from the A420 is implemented. On the assumption that each outlet will produce around 40-50 completions annually, we find the assumptions quite conservative, and readily achievable.

We would advise that the faster the development is built out, the quicker demand for bus services will arise. This has significant positive implications for the earlier delivery of higher quality bus services, that also require greatly less interim revenue support funding.

The delivery of key community facilities is tied to the following programme:

- Primary school extension will be delivered during in phase 1 (PSE site by 300th occupation)
- Primary School in phase 2 (PS by 700th occupation)
- Local Centre retail/community/office/café/restaurant provision in Phase 2 "Rowborough Central" which is envisaged to be in place by 2030
- An additional mixed use centre in "Rowborough East" phase 3, associated with the major formal sports and leisure provision, but which may additionally include some local retail or other uses.
- Early Years provision will be made in the Local Centre

Within the site, the delivery of the spine road that will effect the provision of a bus route, or routes. The proposed phasing is explicit that:

"South Marston Main Street is being delivered ahead of the trigger identified in the relevant outline condition, due to its initial role as a construction access to support the mobilisation of the site and delivery of Phase 1. The provision of access along the Primary Street in Rowborough (Phase 2) will be implemented progressively to support the delivery of development parcels." (3.12)

This means that until sufficient primary streets have been delivered in Rowborough, east of Vicarage Lane, to allow a bus to reach the village route and terminate within it, no bus service will be achievable. We have significant concerns that sub-phases of delivery of streets within Rowborough Central the largest element of the scheme by far (1470 dwellings), will make it in practical terms impossible to provide a bus service to Rowborough for a very long time.

This concern is aggravated by the relatively late s106 trigger for the delivery of the new vehicular connection to the A420 from Rowborough to the P+R site, which is 1200 occupations of 1470 within Rowbrough Village.

This suggests that the only practical way to serve Rowborough with a bus route, prior to that point, will be from Whalebridge using the South Marston Primary Street. This serves phase 1 and phase 4.

Section 4.10 indicates that the full length of the South Martson Primary Street will not be available to general traffic (or buses) until, at the earliest, 300 occupations. Thus, Phase 1 will not benefit from a bus service at the point occupations are taking place. Given only about 260 dwellings are involved, this should be considered a show-stopper.

In partial mitigation, it would be possible to use the existing crossing of the railway to reach the existing S6 service from those initial occupations, within about 900-1200m. This is far from ideal but given the constraints, unavoidable. However the Condition 31 "Footpath 5" link in our view needs to be accelerated, so something safe is available prior to the 300th occupation. This need not be a fully paved ped and cycle link. Providing the link to sub-base level (type 1 aggregate and perhaps temporary scalpings layer) to allow

those who need a bus service to have some safe access to one would be proportionate. While we recognise that this is not an obligation on the developer, this might reasonably be in place by the 100th occupation.

The separate obligation to have secured an adoption agreement and delivery of the Primary Street to practical completion by the time that 50% of dwellings in the adjoining phase are occupied, in practice means that much of the South Marston Primary Street length may be at base course till near the end of the entire build-out. If this length is the main haul route and construction access for that duration, this will be a far from ideal bus operating environment, for a potentially very lengthy period of time. However bus operation is probably achievable, with close collaboration with the Consortium, if they are so willing.

Based on the developers submission, the bus routing and access strategy implied by the proposed phasing in effect should be considered to have three critical stages:

- The phase 1 access, using the South Marston Primary Street, and the initial length of the Phase 2 Primary Street to Rowborough east of Vicarage Lane, to a temporary terminus in Rowborough.
- The completion of a substantial and rational succeeding length/s of street to accommodate a bus route within Rowborough will allow the service to conveniently and safely offer access to services within a reasonable walk of active residential development parcels on which occupations are taking place, before the eastern vehicular connection to the A420 is provided.
- The delivery of a primary street linking from east to west along the entire extent of the development to the Park and Ride site, is an important matter, and greatly more so now than when the application was submitted and determined. This is because it remains far from clear when, if ever, an express public transport sine can be delivered across the western boundary of Great Stall East to effect the major component of the public transport strategy for NEV. If, as we consider likely, the A420 remains approximately as congested as it is today at peak times between Acorn Bridge and Whalebridge, it ***might*** even prove to be **necessary and rational to divert the inter-urban S6 service through the development.** This is something we explicitly indicated when the application was submitted that we would not do, and which has remained our position to this point. In addition, a Park and Ride offer to the town centre might also find that this is the quickest route available.

Therefore, for buses service to be available suitably early, **this will require at least one and probably two features that allowed buses to turn around, to be provided on the primary streets.** These must feature a geometry that offers a minimum 26m inscribed circle diameter. A temporary feature, involving sacrificial works off the centre line of the spine road, could be acceptable. A permanent feature on the spine road could well be both preferable, and would also be likely to offer an opportunity for the feature to calm general traffic speeds, for example at the main Local Centre in Rowborough.

Use of secondary and tertiary streets on a temporary basis to allow buses to turn around a development block or blocks would inappropriately constrain the design code process and demand the extension of standards and geometries to these side streets on the one hand, which is unlikely to be acceptable to the Planning Authority of the applicant, or present unacceptable operational and amenity impacts on the other. Such a solution is therefore unacceptable.

We would strongly advise the applicant and the Council to discuss the appropriate design response to provide these turning facilities with the applicant well prior to submission of the Neighbourhood Design Code for specific phases.

We would highlight at this stage that a good many homes in phase 2 and phase 3 (Rowborough Central and East) are likely to stand well over 400m from the nearest bus stops that can be delivered on this bus route, even if only temporarily. This can be mitigated to a material degree by the optimal siting of bus stops at the spine road junction within Local Centres/Mixed Use area. Accordingly, we would urge that there is early discussion with us well prior to the formulation and submission of the Neighbourhood Design Codes for Phase 2 and 3.

For all phases, the siting of bus stops that optimally relate to strategic pedestrian infrastructure between those phases and the route, will be of the essence. We advise specifically on this matter later in this response. Irrespective, prior to finalising the submission of the Design Code for any element of phase 2 and the design of the reserved matters application for this infrastructure, we would urge that this is very carefully considered in collaboration with ourselves.

3. Interim and ultimate bus services to Rowborough

The phasing strategy must have regard to the delivery of key connectivity and penetration for sustainable modes, including public transport, in line with the expectations in the Local Plan and the supporting SPD policy.

At the time of the initial preparation of the planning application for this development, which was nearly a decade ago in 2012-2013, there was no clarity on the sequence or timing either of adjoining developments, or of many key elements of infrastructure. This included the Public Transport Spine running through Great tall, south of the A420. This would accommodate an express bus service running between the Park and Ride provision, near Acorn Bridge. This will be at the intersection of one of the main vehicular accesses to Rowborough, which requires the delivery of a new underpass beneath the Great Western Main Line.

The strategy also hoped at the outset, to leverage the development of the South Marston Village Extension to restore the bus service to the village that operated prior to 2013. This was a limited service that was financially supported by Swindon Borough Council. The entire budget for such services was withdrawn by the Council and has not been reinstated since.

We were clear at the submission of the scheme, that the size of the existing village, even when 500 new homes were added, would be greatly lesser than that necessary to sustain a bus service on a sufficient frequency to be relevant and marketable. We also expressed concern that the actual route proposed through the extension was both excessively circuitous, while also offering convenient walking distances to potential stops from relatively few dwellings, especially in the existing village. We also made clear that we considered that diverting a long-distance inter-urban route (then the 66, now S6) through the site would be inappropriate.

To date we have consistently advised that the most efficient and probably the most attractive way to serve the site in the longer term would be in connection with the Public Transport Spine through Great Stall East and the P+R site. This provides a direct and efficient link straight to the town centre, but also the Secondary School within Great Stall East and the existing and future facilities at the District Centre further West. It would allow the Rowborough service to combine between the P+R site with the inter-urban provision between Oxford and Swindon to create a credibly sustainable 10-minute service frequency from the P+R facility. However, this arrangement would leave the South Marston Extension unserved by bus.

The deliverability of such a solution nevertheless depends firstly on the completion of the express public transport spine through Great Stall, and then the Eastern Vehicular Link into Rowborough across the A420.

The latter will not be effected until after the 1200th occupation in Rowborough Central, which is indicatively anticipated about 2032.

However, there is now considerable doubt about the timing of the public transport spine through Great Stall. We are not aware that the legal challenges surrounding the adoptable connection across the western boundary of Great Stall East have been resolved. Even if it were, this link will not be deliverable, according to the phasing plans for GSE, until after 2032.

In addition, the anticipated westward extension of the express public transport spine across the A419 over a new dedicated bridge into the existing built up area, looks equally doubtful, certainly in the foreseeable future.

The development bus service will therefore be dependent on access via Whalebridge, and the Primary Street through South Marston, for a considerable period.

Were the Old Vicarage Lane link able to accommodate double decks, it might even be possible to consider a diversion of S6 journeys into the scheme, especially if journey time reliability on the A420 through Whalebridge remains as difficult as it is today. However, it cannot. Nor would this option allow initial phases of Rowborough to be served.

Accordingly, a dedicated bus service to Rowborough is apparently unavoidable. This is likely to overlay the S6 between the town centre Greenbridge and Whalebridge, if it is to operate direct. This risks seriously compromising the economics of both the S6 and the new service. The costs of operation would be high –

even a 30-minute service frequency will demand three buses and an operating cost of about £520,000 per annum. The gap with revenue

Another much cheaper alternative would be to extend an existing service terminating in the near vicinity. The only realistic candidate is likely to be Swindon's Bus Co. service 2. This would lead to both an indirect and relatively intricate route to town, and requires the breaking of the current terminal loop, leaving a risk of some parts of Covingham being further from bus stops. However the net additional resource required to reach the development with a half-hourly offer would be a single vehicle – essentially one third of the cost of a dedicated service. While this level and form of provision would be far more affordable and avoid exhausting the development bus service budget vey early – risking the service being entirely withdrawn part-way through the development build-out – we would also stress that it would have relevance only to a core market of essential public transport users, and would not lead to a substantial public transport mode share.

Whatever happens, it is essential that the highest possible quality of infrastructure to support the bus service in the development is provided to give the services the best chance of success.

4. Condition 8: Strategic Design Code for South Marston and Rowborough

6.1.3 Role of the Strategic Code is to inform the location and design principles for all the strategic mobility infrastructure to enable the development of the individual parcels, to be delivered when required by the phasing strategy for the site."

The SPD covering NEV refers to "Express Bus Services". This specifically involves the main east-west spine that was going to be delivered between the existing town, across a new bridge over the A419 and through Great Stall Village and the District Centre to a park and ride site at Acorn Bridge. As we outline above, this express bus route is not, as we understand it, likely to be put in place in the foreseeable future.

Irrespective, as far as things are now foreseeable, **the primary street through South Marston Extension and Rowborough will provide a very considerable proportion of length of the main and quite possibly the only bus route that is likely to serve the development.**

Therefore, it needs to be emphasised that for any bus route to begin to work, especially having regard to the distance of Rowborough from major local facilities and the town centre, **the most efficient, smooth and direct "level of service"** needs to be provided to bus operations, irrespective of the operator, of kind of service. A one-way peak journey from Rowborough to the centre of town needs to take no more than 35 minutes, and ideally under 30 minutes. One of the major challenges of providing bus services to the Northern Development Area is that the streets available to provide bus services do not meet any of these criteria.

4.1. Primary Street Design Parameters

With regard to the design parameters for the primary street, we would observe and advise as follows:

- We note the intention that Primary Streets should be designed in accordance with Manual for Streets (6.5.3). Manual for Streets has a curious status. Aspects of it are problematic and it led to the publication of Manual for Streets 2, which is neither a Government publication nor does it hold any kind of regulatory force. Both are to be replaced, by the end of 2022, with a "**Manual for Streets 3**", which will be a Government document, though the regulatory weight attachable to it is yet to be decided. We would also advise that Primary Streets are designed in accordance with parameters set out in the **National Model Design Code** which covers well the crucial interface between built form and movement routes, of all kinds. This sets clear expectations that Government intends to be the "starting point" for urban design.
- **6.75m minimum width** exceeds the parameters set out in current CIHT Guidance "Buses and Urban Developments" (2018), and our own urban design recommendations, which advises 6.5m. (6.4.2) Carriageway width of 6.75m will encourage on-street parking on both sides of the carriageway. Rather than allowing for buses to safely pass in both directions, it is more likely to lead to very long lengths of street operating as single carriageways. While driveway crossovers and side roads might offer reasonably regular places for cars to pass, these will be insufficient for buses, and in any event, such an arrangement cannot permit efficient bus operation. Where segregated in-line

cycle provision is made on a primary street, of 2m wide in each direction in line with LTN 01/20, it would be appropriate to narrow the carriageway further to 6.2m, allowing for some widening on tight ends. We cover this later in this response

- **Passive design features to calm traffic** should be employed, including the use of street streets on both sides to crease visual friction.
- We therefore support, **occasional narrowing of the carriageway** for a short length to limit vehicles speed and facilitate safe crossings at key locations. (6.4.3) We would advise that 6-6.1m ought to provide an adequate width, commensurate with the intention of these features, rather than 6.4m, which would not, especially where general traffic is concerned. A length of no more than 10m is recommended. As indicated in 6.5.14, these could incorporate rumble strips and/or changes in surface colour/treatment (such as light coloured stone chips). We would also support median features, and would invite consideration of some of the ideas proposed for Lotmead.
- We note from the schematics on pages 84-85 that vertical deflection features are implicitly under consideration. They are not mentioned in the text. We also note the comments related to **"Key Places"** at section 7.5 of the SDC. We would urge great restraint on this point. If incorporation of active **traffic calming** features is considered absolutely necessary, a 75mm maximum height, and 1:12 maximum transition gradient should be used, in line with Stagecoach published guidance. Such features should be used sparingly. The ambient noise impact of ramps and tables should not be underestimated especially where large vehicles are concerned. Features higher/steeper than this can cause serious health and safety risks for bus passengers especially those standing to prepare to alight, near the back of vehicles, as well as disproportionate discomfort to bus passengers more generally.
- **Cycle provision** is anticipated along the length of the primary streets. *"provision for people cycling will be delivered either as a 3m dedicated cycle track combined to a 2m footway, or a 3m shared pedestrian and cycle way where more direct dedicated cycle routes are provided within the open space"*. While this does broadly comply with **LTN 01/20** standards, especially if it anticipated that fewer than 300 peds/cycles per hour would use the provision, this is not in our view either desirable or expedient. **A much higher level of service to cycles and clear segregation for pedestrians could be better provided for with continuous *2m wide stepped in-line cycle provision***. This would be deliverable within the same or lesser corridor width as that proposed, especially if it accepted that the carriageway width can reduce to a minimum 6.2m. **In-line provision is considered much more desirable by LTN01/20**, not least because the nature and complexity of conflicts between cycles and other road users on 2-way offset side-slung provision is very much greater. This includes at bus stops. In-line provision is also a great deal more legible to all road users, according with a key foundational principle of LTN/01/20. The interface between bus stops and cycle provision is a critical safety issue. Adults cycling and riding on footway are extremely hazardous in any event (hence, generally, it is illegal) and greatly diminishes the real and perceived safety of the pedestrian realm. On a street of this status on a development of this size, we think it appropriate to consider in-line with-flow segregated provision for cycles. This has substantial implications for bus stop design, which would "float" between a cycle track and the vehicular carriageway.
- The approach to be taken to **frontage access** is crucial. However the SDC text is vague on this point. It states *"Some direct access off the primary street will be allowed to individual dwellings or a group dwellings. (6.5.6)"* More clarity is needed. However the diagram on page 104 is rather more illuminating. We welcome the indication that unlimited individual driveway crossovers will not be pursued on primary streets. We would strongly recommend that the presumption is made explicit, that "groups of dwellings" should share frontage access points on primary streets, to rationalise crossings of the cycle provision, and ensure that cars enter the Primary Street in forward gear, which is appropriate given its greater movement function.
- Section A-A on Page 83 summarises the key parameters for the primary street through the built village form. We note that built frontage is anticipated to **stand off as little as 5.5m** from kerblines. While these are residential primary streets that accommodates only development traffic this is below our minimum recommendation of 6m. Here, and in line with the National Model Design Code, we wold recommend that a **minimum of about 9m** is pursued, not just for amenity reasons but to

accommodate an appropriate treatment of parking and frontage access using shared private drives, allowing vehicles to enter the highway in forward gear and better accommodate delivery vehicles.

- The approach to off-curtilage parking on the Primary Street is crucial, as our own guidance makes clear. Here too, the Strategic Design Code is inappropriately vague: *"Occasional parallel visitor parking is allowed."* (6.5.7) We would urge that the ***maximum appropriate use*** is made of parallel **parking bays to accommodate on-street parking** in line with SBC standards. The primary street should not generally permit on-carriageway parking. This suggests a relatively high number of bays will be appropriate, consistent with other design objectives and the incorporation of street trees. As indicated above, the approach to driveway crossovers will be crucial. There is a clash between the desire to have the maximum number of individual frontage accesses, parking bays and street trees.
- We note from the Drainage Strategy Concept Plan on page 67 at section 5 that **swales** are not anticipated to be incorporated on the primary street, certainly within the built development. However this is contradicted at 6.5.9 where this matter is left explicitly open. *"Verges are to be included as part of the street section, to also include trees and swales where appropriate"*. This matter has a number of significant implications for more detailed design and we would expect those lengths of primary street that are intended to incorporate swales to be more clearly indicated. **Where swales are present this presents a major constraint** for the location and design of crossovers, bus stops, and parking bays, as well as presenting issues for casual pedestrian crossing (including for parcel delivery) and the way in which cycle provision is made. We can see that **provision of swales outside the village built up areas would be more appropriate**, along "rural and "parkland" sections in which case, two-way cycle provision might be more appropriate off-set from the carriageway. The points where the transition is made between village and rural sections are made would then involve cyclists crossing the street in one direction, however, which is not ideal.

4.2. Bus stops provision and specification

The Strategic Design Code does not go much beyond setting the "five minute walk" standard that is reflects an uncritical application of a standard that was never evidenced, and no longer exists. The IHT guidance of 2000 that refers to this standard (and a 400m maximum walking distance" was superseded in 2018, in the CIHT publication "Buses and Urban Developments".

The Local Plan and SPD makes reference to a **need to achieve mode shift away** from private car use on all the strategic allocations including NEV. A much higher mode share for bus is therefore demanded by policy. This requirement demands buses to be much more conveniently accessible than the SDC makes provision for.

CIHT expects 300m to be used as a starting point. In urban contexts we broadly agree, with the proviso that a degree of flexibility is needed to account for context and urban form. We consider that 400m is a backstop threshold where bus routes are intended to penetrate development areas, as is the case here. We also recognise that some small parts of the development – notably the eastern end of Phase 3 – might not be within a 400m walk of stops. The key is to provide a rational efficient built form incorporating a rational and efficient bus route and thus slavish pursuit of a threshold that would result in a contrived and unsustainable solutions should be avoided.

As we have already said, we had anticipated that the secondary road referred to as **Secondary Street Phase B** would actually be part of the primary movement network. The most logical means of providing a route to Rowborough on the assumption that the express bus route were deliverable along the Public Transport Spine to the P+R near acorn bridge, would involve extending that service across the A420 into Rowborough, with the bus using that loop to circulate around the whole village. The combination of demands from Rowborough, the P+R and Great Stall East will be absolutely crucial to credibly support a bus service at a frequency that will be marketable and effective on a financially sustainable basis.

However, it is unclear when or even if, this service pattern could be achieved. Nevertheless both obvious alternative routes would benefit from using this same terminal loop:

- An extension of the Lotmead service north through GSE to the P+R and terminating at Rowborough (having used the Southern Connector Road to approach NEV from Commonhead).

- Running out as envisaged by the applicant and as covered earlier in this response, via the South Marston Village Extension.

All that said, any threshold needs to have regard to actual walking routes, not simplistic “crows flight” radii. The relationship of bus stop sites to the built form, legible and pleasant direct pedestrian corridors and key facilities and local centres in particular, is critical, as our published advice makes clear.

It is evident that the development will consist of two very distinct and physically separate villages. Buses will pass through significant areas of undeveloped land to and between them. As our guidance makes plain there is little sense in introducing bus stops on these sections of primary street. Spacing of bus stops will extend between the South Marston Extension and Rowborough Villages.

As an aside, the villages both stand off the rail line a considerable distance, so walking or cycling trips off-site to existing inter-urban services on the A420 are that much more extended.

We welcome that some attempt has been made to indicate bus stops within the SDC, at section . This assumes that buses will circulate largely through the southern parts of the villages, especially Rowborough Central. The bus route physically stands off from much of the housing in South Marston Extension – a concern we have consistently raised. Notwithstanding what is indicated, in the light of our published advice and the commentary above we would advise as follows, following the route through Character Areas set out in Section 8 of the SDC from west to east:

- Manor Green and South Marston Village Centre: a pair of stops should be sited just west of the eastern junction of the loop road, within the developed area. A second pair might be necessary further west where key pedestrian and cycle links cross the Primary Street, though these will be some distance from the nearest dwellings.
- Marston Gate and Nightingale Corner: We have already commented on the Neighbourhood Desgn Code. We recommend that the bus stops are repositioned slightly to optimise hinterlands and safety. A second pair near the junction with Old Vicarage Lane is probably justifiable to optimise convenient access without an excessive proliferation of stops.
- Valley View and Neighbourhood Hub: This is one of the most important parts of the development. It incorporates a considerable length of bus route as well as being a key focus for amenities. The community hub should be designed to maximise the potential to help facilitate inter-modal journeys, since a local centre typically represents a point where change of mode can take advantage of local retail, early years and the primary school. For example, walking or cycling journeys to the community hub to drop of children, can be combined with an onward bus journey if suitable secure and surveilled parking facilities are available for cycles. As well as the stops already anticipated for the Local Centre, an additional pair would be justified where the Primary Street enters the Village. To the east of the Local Centre, the next pair of stops should relate to the junction with Secondary Route Phase B, which is an obvious node. Stops should be downstream of each arm of this junction, as this maximises routing flexibility without introducing an excess of facilities, as well as being generally safer. We discuss the routing of buses on a route incorporating the Secondary Street Loop elsewhere. A bus turning facility should be provided at the local centre which might be temporary. It is also important that facilities are provided to allow buses to wait time without blocking passing traffic. These stops are likely to be the timing point for services and lay-bys or 2.m wide partial pull-ins should be provided, as early running buses may have to wait time.
- Nightingale Edge: This represents the northern portion of “Rowborough Central” which is the focus for the residential quantum. As we raise earlier, we had anticipated the loop street through this area to form part of the Primary Street network to accommodate a bus route, circulating in a single direction; of which anti-clockwise is probably very slightly preferable. If a stop is sited at the eastern end of the loop street as referred to previously, this suggests that probably two further stops, about 250-330m apart, would be justified. Siting where key walking and cycling routes, and/or the green corridors intersect the street, offer ample good opportunites. This will bring a very large proportion of homes well within a 300m walking distance of the bus service, without an excessive number of stops. This is very important if a higher bus mode share is to be achieved, as Council policy for NEV seeks.

- Rowborough Gate: the Primary Street is offset very considerably to the west, with development extending a good distance to the east. Two pairs of stops will be needed. It would be logical for one to be closely related to the mixed use hub, albeit probably slightly to the north, as this would then also pick up the natural node formed by the junction with the secondary street. The second pair would be further north. It is hoped that a pedestrian route will be provided running along the northern edge of the residential area, south of Nightingale Wood, and the intersection of this would be a logical place to site stops so that already long walks to the bus route from the far eastern end of this area are not further extended by needless circuity. As site slightly further north at the northern edge of the developed area would also be a good choice.

All bus stops should benefit from a free-standing flag with integral timetable case, consistent with the adopting authority's emerging standards for quality provision, which has yet to be determined. Shelters are especially important at the Local Centres and at least one, town centre-bound, is no doubt justified within South Marston Extension.

We recommend that early discussion is had with the Local Highways and Transport Authority with respect to real-time information displays being provided at most stops, as this will eed to reflect a consistent level of provision across New Eastern Villages.

4.3. Frontage Treatment on bus routes

We find that the relationship between built form and the street has a very wide range of direct and indirect impacts on the safe and efficient operation of bus services, and on wider residential amenity, as well as the experience of the place.

The Strategic Design Code is relatively high level and does not seek to make clear statements about the more detailed design of main street frontages. Rather "*Each relevant Character Area Design Code will identify the unifying elements and the range of variations allowed for each frontage typology, to inform the character of each street." (7.2.9)*

This is in our view regrettable, as this should be set at this stage to act as a "fix" for subsequent Neighbourhood Design Coding and Reserved Matters. The approach taken by the developers at Lotmead to establishing these parameters in more detail at this stage, is one that we would highlight as being greatly preferable.

The approach to the Primary Street needs to start with block structure and the density of intersecting vehicular side streets. We recommend that a lesser number of side roads intersects the primary street not least to enhance the effectiveness of segregated cycle provision. It also reduces the incidence of turning movements in and out of side streets. Right turning movement are likely to block the primary street frequently.

The frontage treatment of Primary Streets will be crucial to the legibility of the place as well as place-making and development character. However, the submission says relatively little about this. We recognise that the density and form of the development will be relatively relaxed – typically less than 40 dwellings per Hectare even along the Primary Street, where densities are to be significantly higher.

We would strongly urge as follows:

- Front parking directly accessed from the Primary Street should ***not*** be provided along these frontages.
- **Front gardens or privacy strips should be deeper**, incorporating features such as railings or with boundary hedge along these frontages. This will among other things, help establish a clear sense of place and better legibility. It creates an appropriate standoff from a busier carriageway, for amenity reasons. It also allows for an appropriate sight line to be created where private drives cross the strategic cycle provision.
- Unlimited **individual private driveway** crossovers should not be permitted on Primary Street Frontages. **"Side parking" is especially problematic on a busy primary street like this**. This necessitates a very frequent use of driveway crossovers to individual properties: a pair every 11-

13m. Driveways will be single width and minimum 6m long, providing only one parking space per plot on curtilage. Garages should not be counted as parking spaces as they are almost never used as such. A second or third car will therefore need to be parked elsewhere. The length of kerbline between each crossover on typical semi-detached housetypes is ideal to allow the second car to be left on the carriageway in front of each dwelling. Because residents do not care to move cars around on double length "tandem" drives, even where a longer drive is provided, as it typically would be on a primary street, this leads to **continuous on-carriageway parking**, unless alternative off-carriageway provision is made. This has a particularly serious impact on the ability of bus services to effectively serve developments. In addition, where cars are parked on drives, these typically reverse out onto the highway, crossing footways and cycle provision as they do so. It is impossible to provide adequate sight lines, especially here where dedicated cycle provision is anticipated along the primary route, as these can be moving at considerable speeds. Cars should be entering the highway in forward gear, and individual drives do not readily assure this.

- Parking should not generally be permitted on carriageway. We observe that SBC policy is to seek parallel parking bays, 6m long and 2.5m wide, with small tapers to accommodate on-street parking. However with 13m or so between crossovers that the proposals entail, what this means in practice is that a pair of bays between crossovers would not permit space for tree pits. Certainly a single bay could be installed between each crossover, with trees at either end, which would start to resolve this tension. However in our experience, this will not permit the provision of sufficient unallocated/visitor parking to meet standards. Even if parking does not burden the primary street carriageway, it start to put severe pressure on side streets, including high incidence of pavement parking on both sides, which we would expect the Local Highways Authority to resist.

In summary then we would urge as follows:

- Homes are set further back from the kerbline, as anticipated by NMDC, creating more room to accommodate street trees, segregated cycle provision, and rationalised frontage access.
- Driveways permit at least two cars to be parked on curtilage per plot (net of Garage)
- Crossovers are rationalised through the use of shared private drives, allowing entry to the primary street in forward gear, and providing more scope for both street trees and parking bays.

5. Concluding Comments

Both with respect to the Discharge of Condition 6 and 8 we conclude with regret, that the applicant has failed to provide sufficient information to demonstrate how bus services will be facilitated in such a way that the intent of adopted policies in the Local Plan and the SPD for New Eastern Villages can demonstrably be achieved. In terms of the phased provision of bus route and stopping facilities, the submission is especially vague, despite the explicit requirements of the Conditions.

We have set out above our advice on how the phasing and overall design parameters for the development to be set in the Strategic Design Code, can properly address these matters.

We therefore urge the applicant and the Council to work collaboratively with us to address these issues, which actually have much broader implications for all modes, and the experience of the place by residents and many others visiting and passing through.

We must emphasise that if these matters are not appropriately tackled, the Council's aspirations for the delivery of a high standard of bus service in Rowborough will be achieved in the foreseeable future, if ever. The quality of bus service that would credibly achieve a mode shift from private cars, as the Local Plan, LTP and SPD policy for the allocation explicitly set out to achieve, would be fatally compromised.

We trust that this response is helpful to the applicant, you and your colleagues in the Council's Highways Development Management function, who are copied in. Naturally, all parties should feel welcome to contact the undersigned in the first instance, to discuss the matters we raise. On behalf of Stagecoach West and Swindon's Bus Company, I am

Yours sincerely,

Nick Small

Head of Strategic Development and the Built Environment